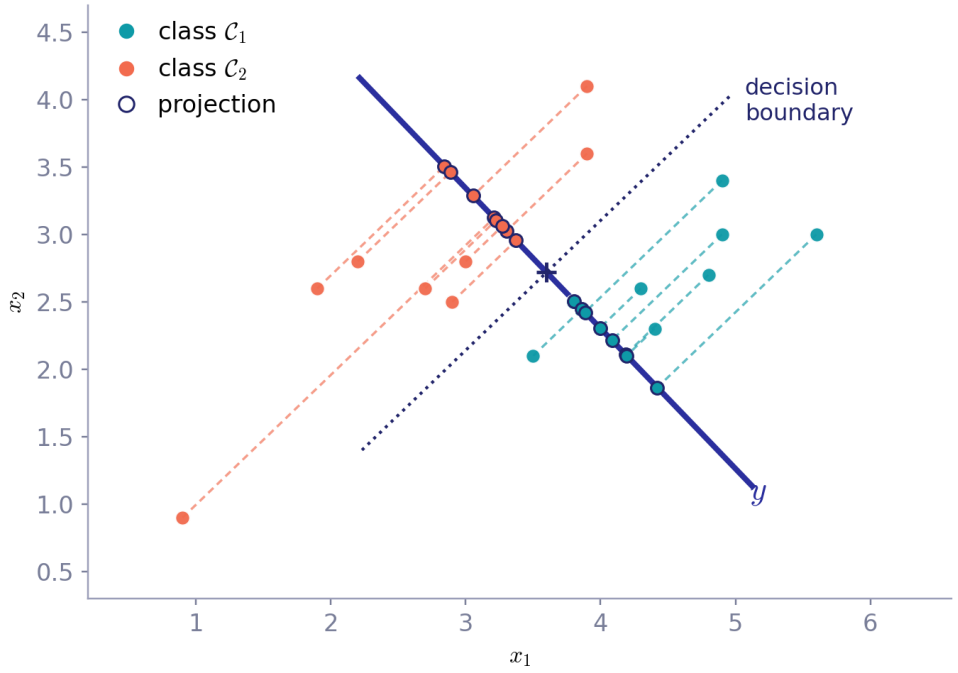
Fisher's Linear Discriminant



Two classes in 2D, projected onto Fisher's direction

Machine Learning · MSc in Computational and Applied Mathematics
UC3M

The goal: classify by projecting to one dimension

1 Project

Collapse every input vector x to a single number.

$$y = \mathbf{w}^t \mathbf{x}$$

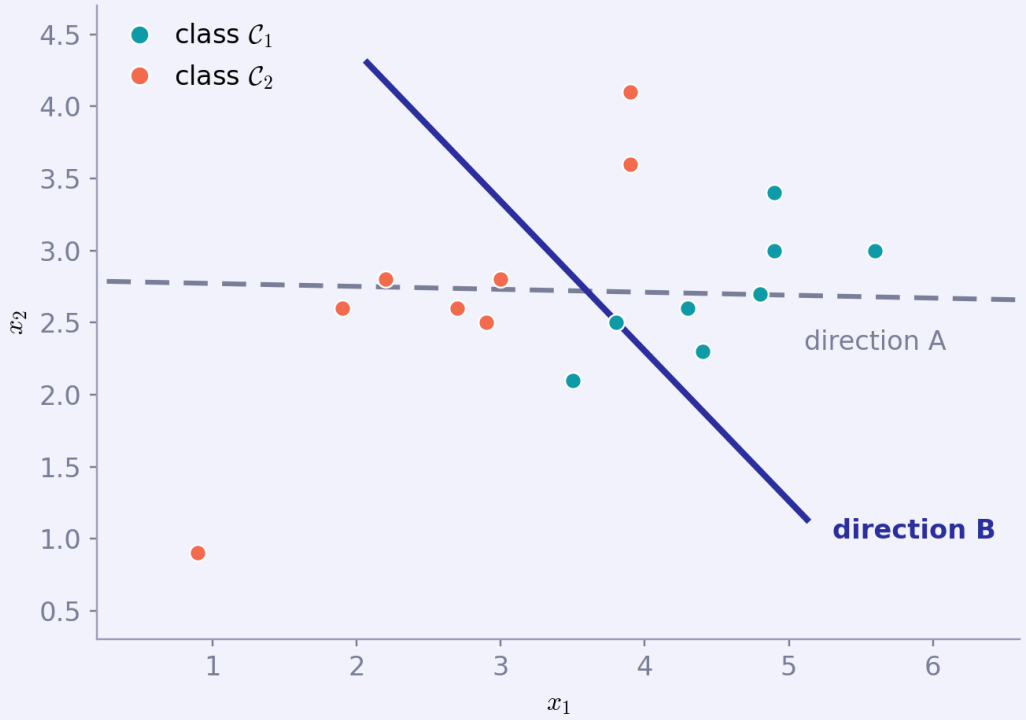
2 Threshold

Pick y_0 and assign the class from the sign of $y - y_0$.

$$\mathbf{x} \mapsto \mathcal{C}_1 \text{ if } y \geq y_0, \qquad \mathcal{C}_2 \text{ otherwise}$$

3 Find the optimal w

Each direction loses different information: classes that are apart in the original space can overlap on the line.



Direction A mixes the classes; direction B keeps them apart.

First idea: push the projected means apart

Slides 63-65

Idea

Choose w so the projected class means are as far apart as possible.

$$\mu_2 - \mu_1 = \mathbf{w}^t(\mathbf{m}_2 - \mathbf{m}_1), \qquad \mu_k = \mathbf{w}^t\mathbf{m}_k$$

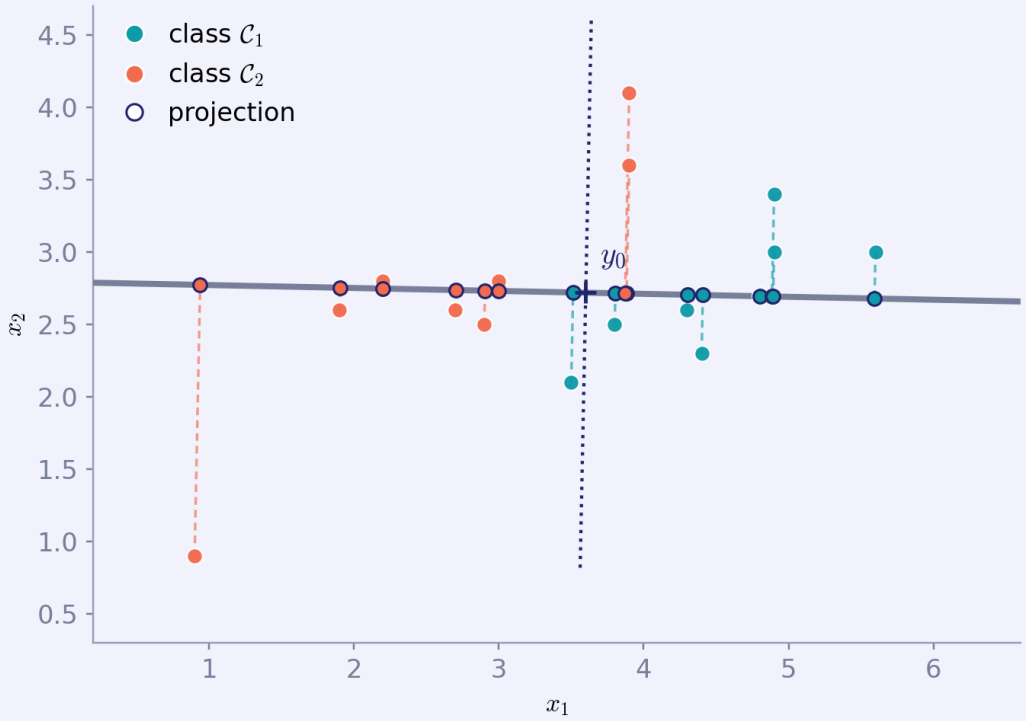
Constraint

Without it, $\mu_2 - \mu_1$ grows with the length of w.

$$\sum_i w_i^2 = 1 \;\Rightarrow\; \mathbf{w} \propto \mathbf{m}_2 - \mathbf{m}_1$$

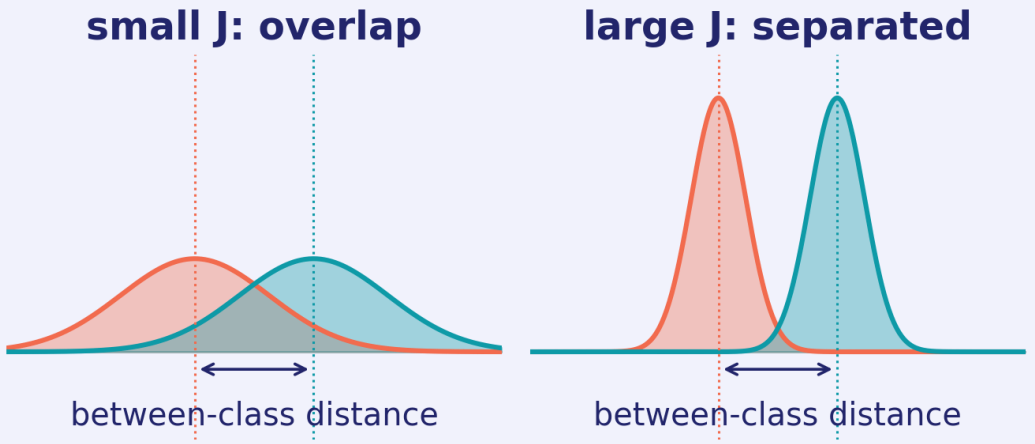
Limitation

It ignores the spread of each class. With correlated features the clouds are elongated and the projections overlap.



Projecting on $m_1 - m_2$: 3 of 16 points end up on the wrong side of y_0.

Fisher's criterion: far-apart means, tight classes



Same means in both panels; only the within-class spread changes.

Between classes

Distance of the projected means: $(\mu_2 - \mu_1)^2$. Make it large.

Within classes

Spread inside each projected class: $s_1^2 + s_2^2$. Make it small.

Fisher criterion

$$J(\mathbf{w}) = \frac{(\mu_2 - \mu_1)^2}{s_1^2 + s_2^2} = \frac{\mathbf{w}^t \mathbf{S}_B \mathbf{w}}{\mathbf{w}^t \mathbf{S}_W \mathbf{w}}$$

Within-class variance after projection

$$s_k^2 = \sum_{n \in \mathcal{C}_k} (y_n - \mu_k)^2, \qquad y_n = \mathbf{w}^t \mathbf{x}_n$$

Scatter matrices in the original space

$$\mathbf{S}_B = (\mathbf{m}_2 - \mathbf{m}_1)(\mathbf{m}_2 - \mathbf{m}_1)^t$$

$$\mathbf{S}_W = \sum_{n \in \mathcal{C}_1} (\mathbf{x}_n - \mathbf{m}_1)(\mathbf{x}_n - \mathbf{m}_1)^t + \sum_{n \in \mathcal{C}_2} (\mathbf{x}_n - \mathbf{m}_2)(\mathbf{x}_n - \mathbf{m}_2)^t$$

Maximising J(w): the optimal direction

1 **Differentiate J(w) and set it to zero**

$$\nabla_{\mathbf{w}} J = 0 \;\Rightarrow\; (\mathbf{w}^t \mathbf{S}_B \mathbf{w})\, \mathbf{S}_W \mathbf{w} = (\mathbf{w}^t \mathbf{S}_W \mathbf{w})\, \mathbf{S}_B \mathbf{w}$$

2 **$S_B w$ always points along $m_2 - m_1$**

$$\mathbf{S}_B \mathbf{w} = (\mathbf{m}_2 - \mathbf{m}_1)\,[(\mathbf{m}_2 - \mathbf{m}_1)^t \mathbf{w}] \;\parallel\; \mathbf{m}_2 - \mathbf{m}_1$$

3 **Only the direction matters**

Drop the scalar factors and multiply both sides by S_W^{-1}.

Fisher's direction

$$\mathbf{w} \;\propto\; \mathbf{S}_W^{-1}(\mathbf{m}_1 - \mathbf{m}_2)$$

If $S_W \propto I$, this reduces to the mean-difference direction:

$$\mathbf{S}_W \propto \mathbf{I} \;\Rightarrow\; \mathbf{w} \propto \mathbf{m}_1 - \mathbf{m}_2$$

Threshold: midpoint of the projected means

$$y_0 = \mathbf{w}^t \frac{\mathbf{m}_1 + \mathbf{m}_2}{2}$$

The slides write $m_2 - m_1$; flipping the sign only swaps the side of C_1.

Toy example: two classes in 2D

Running example

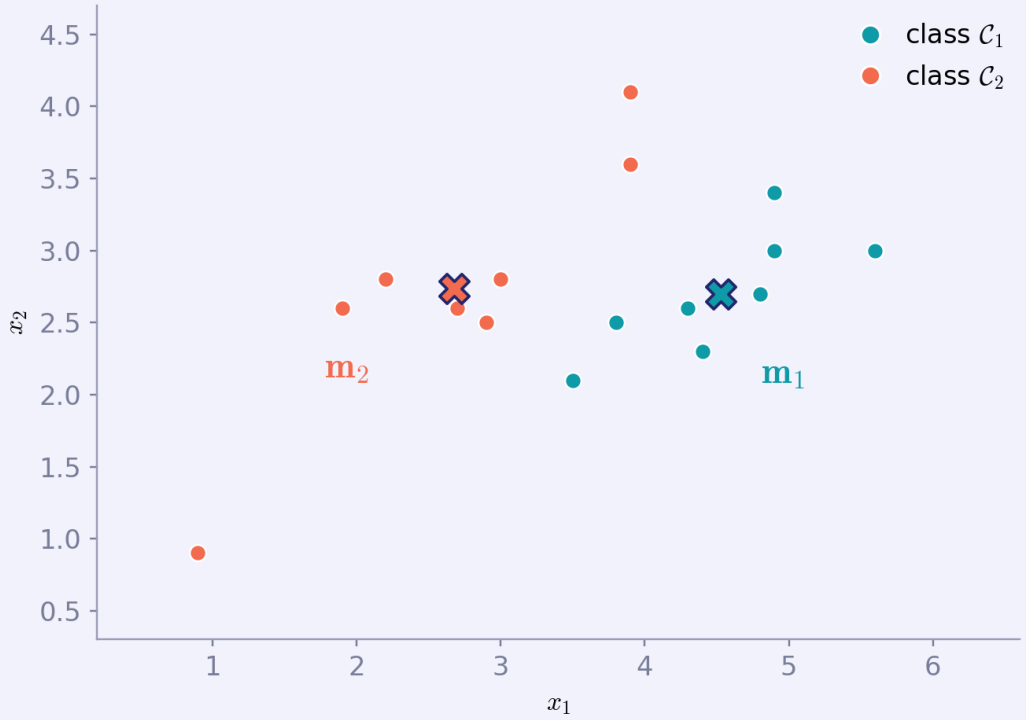
C_1	x_1	x_2
1	4.8	2.7
2	3.8	2.5
3	4.3	2.6
4	3.5	2.1
5	4.9	3.4
6	4.4	2.3
7	4.9	3.0
8	5.6	3.0

C_2	x_1	x_2
1	1.9	2.6
2	2.2	2.8
3	2.9	2.5
4	3.9	3.6
5	0.9	0.9
6	3.9	4.1
7	2.7	2.6
8	3.0	2.8

Class means

$$\mathbf{m}_1 = (4.53,\ 2.70)^t, \qquad \mathbf{m}_2 = (2.68,\ 2.74)^t$$

The clouds are elongated along the diagonal, so x_1 alone is a poor direction.



Crosses mark the class means m_1 and m_2.

Toy example: computing Fisher's direction

Running example

Within-class scatter and its inverse

$$\mathbf{S}_W = \begin{bmatrix} 10.29 & 7.56 \\ 7.56 & 7.32 \end{bmatrix} \qquad \mathbf{S}_W^{-1} = \begin{bmatrix} 0.402 & -0.415 \\ -0.415 & 0.566 \end{bmatrix}$$

Difference of means

$$\mathbf{m}_1 - \mathbf{m}_2 = \begin{bmatrix} 1.85 \\ -0.04 \end{bmatrix}$$

Fisher's direction

$$\mathbf{w} \propto \mathbf{S}_W^{-1}(\mathbf{m}_1 - \mathbf{m}_2) = \begin{bmatrix} 0.76 \\ -0.79 \end{bmatrix} \rightarrow \hat{\mathbf{w}} = (0.69, \ -0.72)^t$$

Fisher criterion J(w)

1.44

Fisher direction

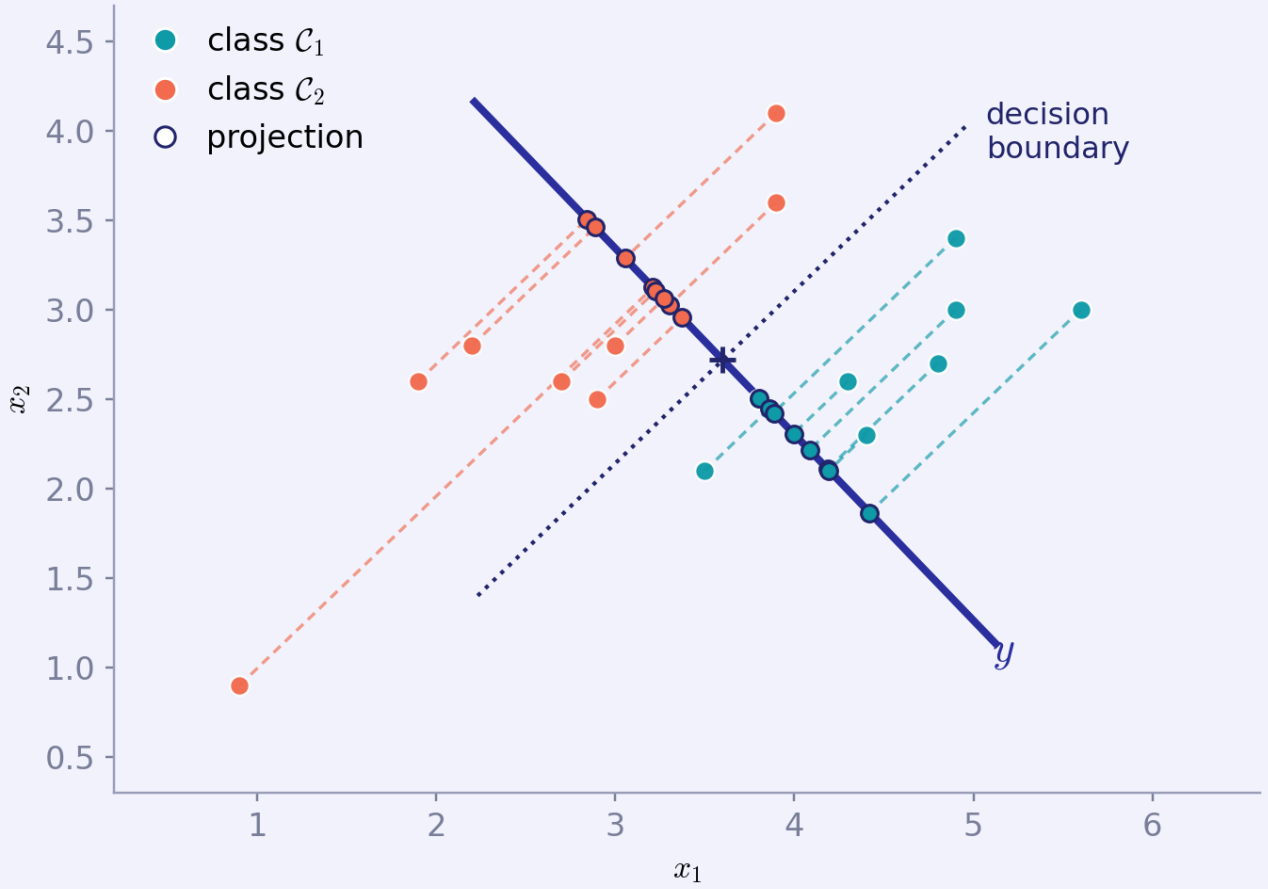
0.34

Mean-difference direction

About 4 times larger

Toy example: each point projected onto the line

Running example



Projected values $y = \hat{w}^t x$

- C_1: 0.83 … 1.72 (mean 1.19)
- C_2: −0.56 … 0.21 (mean −0.12)

Threshold

$$y_0 = \mathbf{w}^t \frac{\mathbf{m}_1 + \mathbf{m}_2}{2}$$ = 0.54, between the two groups

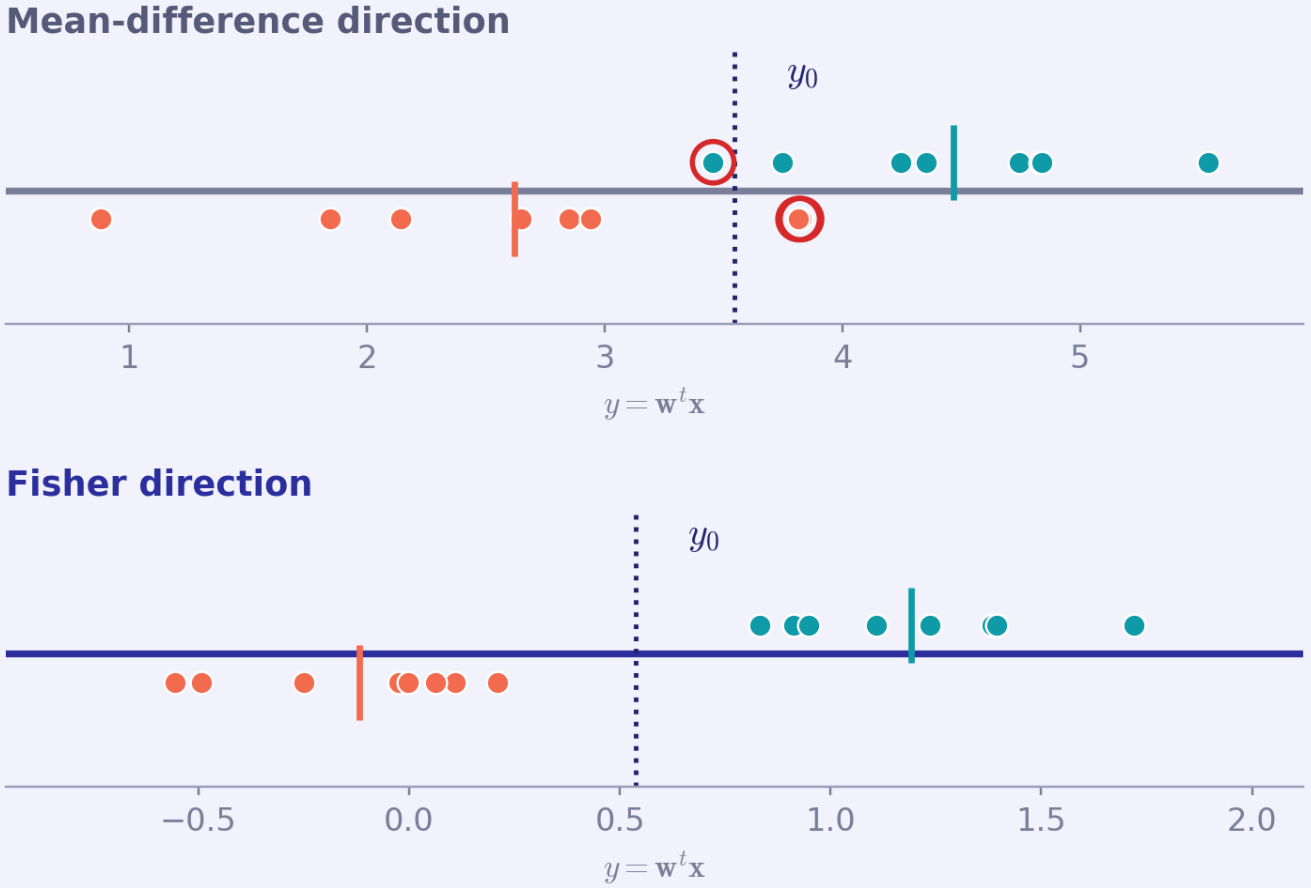
Result

16 / 16

points correctly classified: every C_1 point has $y > y_0$, every C_2 point has $y < y_0$.

Same data, two one-dimensional views

Running example



Mean-difference direction

3 / 16 misclassified

J = 0.34

Fisher direction

0 / 16 misclassified

J = 1.44

Red circles mark errors; two C_2 points almost coincide near y ≈ 3.8. Short bars are the projected class means.

Take-aways

1

Project, then threshold

Fisher's discriminant reduces classification to $y = w^t x$ and a cut-off y_0.

2

Maximise J(w)

Far-apart projected means and small within-class variance give $w \propto S_W^{-1}(m_1 - m_2)$.

3

Covariance matters

In the toy example the correction S_W^{-1} lowers the errors from 3 to 0 out of 16.

In R (slide 70)

```
library(MASS)
fit <- lda(class ~ x1 + x2, data = myData)
plot(fit)
predict(fit, newdata)$class
```

Your turn: Fisher's discriminant by hand

Exercise · about 5 min

C_1	x_1	x_2
1	4	3
2	6	1

C_2	x_1	x_2
1	0	3
2	4	1

1. **Class means**
   Compute m_1, m_2 and $m_1 - m_2$.

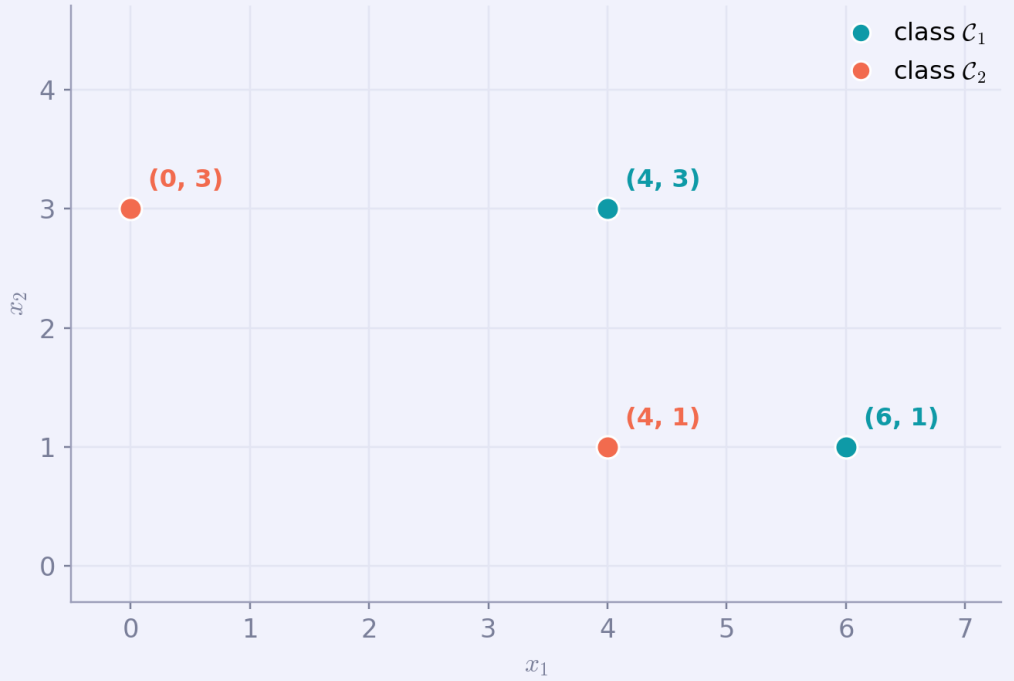
2. **Within-class scatter**
   Centre each class and compute $S_W = S_1 + S_2$.

3. **Invert S_W**
   Use det and the adjugate of a 2×2 matrix.

4. **Fisher's direction**
   $w \propto S_W^{-1}(m_1 - m_2)$; scale it to whole numbers.

5. **Project and check**
   $y = w^t x$ for the 4 points, $y_0 = w^t(m_1+m_2)/2$. Do the classes separate?

Bonus: compute J(w) for your w and for $w = m_1 - m_2$. Which one separates better?



Plot the points, then draw your projection line and threshold.

Hint: for a 2×2 matrix, inverse = (1/det) · [[d, −b], [−c, a]].

Solution on paper

1 **Means**

$$\mathbf{m}_1 = (5,\ 2)^t, \qquad \mathbf{m}_2 = (2,\ 2)^t, \qquad \mathbf{m}_1 - \mathbf{m}_2 = (3,\ 0)^t$$

2 **Scatter: centred points are ±(1, −1) in C_1 and ±(2, −1) in C_2; $S_k = 2vv^t$**

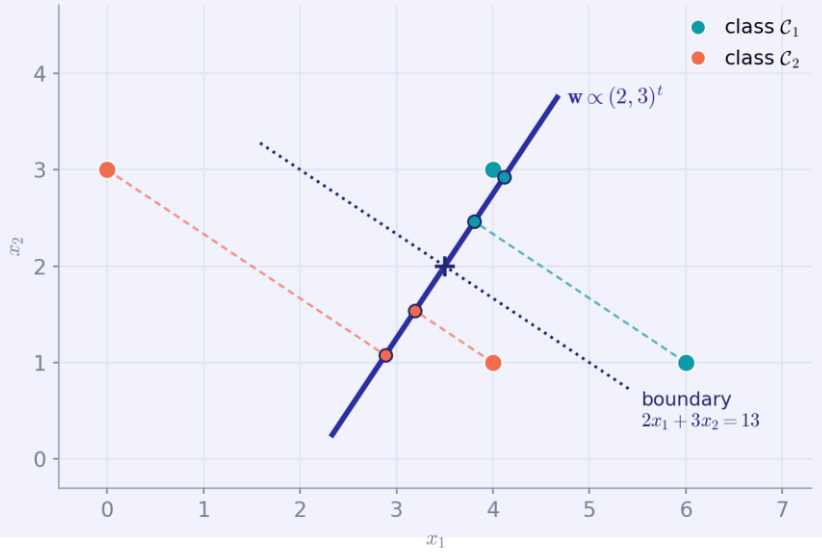
$$\mathbf{S}_1 = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}, \quad \mathbf{S}_2 = \begin{bmatrix} 8 & -4 \\ -4 & 2 \end{bmatrix} \Rightarrow \mathbf{S}_W = \begin{bmatrix} 10 & -6 \\ -6 & 4 \end{bmatrix}$$

3 **Inverse: det $S_W = 10 \cdot 4 - (-6)^2 = 4$**

$$\mathbf{S}_W^{-1} = \frac{1}{4}\begin{bmatrix} 4 & 6 \\ 6 & 10 \end{bmatrix} = \begin{bmatrix} 1 & 1.5 \\ 1.5 & 2.5 \end{bmatrix}$$

4 **Direction**

$$\mathbf{w} \propto \mathbf{S}_W^{-1}(\mathbf{m}_1 - \mathbf{m}_2) = \begin{bmatrix} 1 & 1.5 \\ 1.5 & 2.5 \end{bmatrix}\begin{bmatrix} 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 4.5 \end{bmatrix} \propto \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$



5 **Project and check, $y = 2x_1 + 3x_2$**

C_1: 17, 15 (mean 16) C_2: 9, 11 (mean 10)

$y_0 = 13$, and 15, 17 > 13 > 9, 11: separated.

Bonus: $J(w) = 9$ versus $J(m_1 - m_2) = 0.9$. Projecting on x_1 alone sends (4, 1) to the wrong side.